Ebenezer Isaac

IoT Engineer / Edge AI Researcher / Full-Stack Developer

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Summary

Internet of Things (IoT) engineer with 4+ years of software experience across smart buildings, edge AI and enterprise web platforms. Currently building smart-building systems at Cosysense on Amazon Web Services (AWS) IoT Core and Long Range Wide Area Network (LoRaWAN) sensors. MSc in Systems Engineering for the Internet of Things at UCL, with a Microsoft-supervised dissertation that cut simulated traffic delay by 18.3% using a fine-tuned language model on a 6 GB GPU. Previously almost three years at IBM, delivering IBM.com and SAP support for Volkswagen Financial Services.

Skills

- **IoT and embedded:** AWS IoT Core, LoRaWAN, Arduino, Raspberry Pi
- **Cloud and data:** AWS Lambda, DynamoDB, PostgreSQL, Docker
- **Software:** Python, Node.js, React
- **AI:** Model Context Protocol (MCP), QLoRA fine-tuning
- **Enterprise platforms:** Adobe Experience Manager (AEM), SAP Business Warehouse (BW)
- **Soft skills:** Team leadership, mentoring, stakeholder communication

Work Experience

Cosysense Ltd - IoT Engineer Duration: March 2026 to Present

Automating smart-building client onboarding, from signed deal to installed sensors and energy-savings reports. Tech stack: AWS IoT Core (LoRaWAN), Lambda, DynamoDB, PostgreSQL, Cognito, React, Leaflet

- Mapped 129 undocumented client onboarding steps and identified 80% as automatable, worth about 35 hours of team time.
- Engineered a Model Context Protocol (MCP) gateway that gives AI assistants access to the company's admin platform as the signed-in user, auditing every call and confirming writes first.
- Built tooling that turns a building's documents into client proposals and bills of materials (BOMs) for smart-building deployments.
- Develop remote onboarding and fleet management for LoRaWAN sensors on AWS IoT Core, Lambda and DynamoDB.

Cosysense Ltd - IoT Intern Duration: January 2026 to March 2026

Tech stack: React, Vite, Tailwind CSS, Shadcn, Leaflet

- Led UI and UX development for the internal admin dashboard in a two-person agile team, using React, Tailwind CSS and Shadcn.
- Migrated floor-plan mapping to React Leaflet, engineering a custom coordinate system to place and manage over 20 distinct IoT device contexts.
- Integrated AWS Cognito for identity management, secure credential handling and self-service password resets, strengthening platform security.
- Applied Brick ontology principles to standardise how hardware data points are organised, then translated those IoT data models into clear interfaces.

Jatpoint - Full Stack Application Developer Duration: December 2025 to March 2026

Lead architect for the 8-week end-to-end Phase 2 rebuild of the platform.

- Built location-based attendance checks using radial geofencing, within a Flutter app and Node.js backend deployed on AWS EC2, RDS and S3.
- Set up continuous integration and deployment (CI/CD) pipelines with GitHub Actions and Docker, cutting integration time for staging environments.

IBM India Private Limited - AEM Sites Application Developer Duration: November 2023 to July 2025

Client: IBM.com

- Led roughly 50% of IBM.com's right-to-left (RTL) enablement code changes in a three-person team, proposing alternative components to de-risk timelines.
- Cut issue-triage time by 70% with a Node.js tool mapping dependencies across 240+ Adobe Experience Manager (AEM) components, plus Python automation saving a week of planning.
- Directed the Carbon Design System v2 UI modernisation for an 8-person team, mentoring developers in CSS and Sass and unblocking three complex components.

IBM India Private Limited - SAP BI AMS BW Consultant Duration: September 2022 to October 2023

Client: Volkswagen Financial Services

- Cut average incident first-response time by 80% by introducing Python script automation for Volkswagen Financial Services' SAP BW application support.
- Reduced average ticket closure time by more than 24 hours, and escalated a recurring issue pattern early, creating an 8-hour window that prevented a major priority-one incident.

College of Engineering, Guindy - Full Stack Developer Intern Duration: July 2022 to August 2022

Client: Center for Academic Courses, Anna University

- Reduced manual work and errors by 90% with a PDF generator that templates official letters, then trained university staff to adopt it.

IBM India Private Limited - Associate Application Developer Intern Duration: January 2022 to June 2022

- Earned the SAP S/4HANA 2.0 certification during a six-month internship and gained exposure to a full project lifecycle.

Rishabh Software Private Limited, Vadodara - Machine Learning Intern Duration: January 2020 to June 2020

- Built a time and expense tracker application and prototyped a machine learning model for identifying intoxicated faces.

Freelance - Web Developer Duration: January 2018 to December 2022

Part-time during BCA and MCA

- Built and maintained websites and small web apps for local businesses and community groups, covering requirements, content, basic search engine optimisation (SEO) and hosting.

Projects

- **Edge Negotiator (MSc dissertation, UCL with Microsoft):** Fine-tuned a small language model with QLoRA to control simulated traffic signals on a 6 GB GPU, with a tamper-evident audit log.
- **MycroLinks (MCA dissertation):** A mobile app and NodeMCU (ESP8266) board that add real-time remote control to ordinary electric devices. Built with Flutter, Firebase and Arduino.
- **Fingerprint attendance system (BCA dissertation):** Distributed fingerprint readers run from one admin portal, with audit logs and CSV or Excel exports. Built with Java, Python, Raspberry Pi, Arduino and MySQL.
- **Android password manager:** Encrypts passwords on the phone with keys held in the cloud, and syncs them securely. Built with Google Tink, GCP KMS, Node.js and Firebase.
- **credly-readme-stats:** Embeddable cards that show Credly badges in any GitHub README, with 6 card types and 20 themes. Built with TypeScript.

Education

MSc Systems Engineering for the Internet of Things **Expected 2026**
University College London

Master of Computer Applications (Cyber Security and Internet of Things) **2022**
College of Engineering, Guindy, Anna University

Bachelor of Computer Applications **2020**
Maharaja Sayajirao University of Baroda

Volunteering

- **Volunteer Youth Mentor – Crossroads (2024):** Mentored young people and helped run community events.
- **MCA Class Representative – Anna University (2020 to 2022):** Elected link between the cohort and faculty on timetables, attendance and student welfare.

Certifications

- AWS Certified Cloud Practitioner
- Adobe Certified Expert: AEM Sites Developer
- Adobe Certified Professional: AEM Forms Backend Developer
- SAP Certified Development Specialist: ABAP for SAP HANA 2.0

Additional

Languages: English (IELTS 8.0), Tamil (Native), Hindi (Fluent)
Referees: Available on request